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Files for peer review

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Name	Type of File	Size	Page
SimDec_dashboard_IEEE.pdf	Main Document - PDF	1.3 MB	Page 4
SimDec_dashboard_IEEE_supplementary.pdf	LaTeX Supplementary File	58.7 KB	Page 16

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Global sensitivity analysis made easy through interactive visualization

Pamphile Roy, Mariia Kozlova, Justus Helo, Julian Scott Yeomans, and Andrea Saltelli

Abstract—**One reason for the uneven adoption of global sensitivity analysis methods across various fields is that many approaches are mathematically complex and require specific training. We introduce an all-in-one platform for global sensitivity analysis that, while based on mathematical techniques, is visual and easy to understand, thereby minimizing the learning curve for users. The method and the associated web-based, open-access tool are demonstrated through four case studies involving (i) urban dynamics, (ii) heat exchange engineering systems, (iii) corporate investment, and (iv) a medical application to breast cancer diagnosis. The cases cover different domains and problem types—(i) independent input, (ii) dependent input, (iii) multi-criteria decision analysis, and (iv) given input—and the study aims to promote a wider adoption of this new approach.**

Index Terms—**Global sensitivity analysis, SimDec, Sobol' indices, data visualization**

I. Introduction

TWO advanced research streams address sensitivity analysis from complementary perspectives. Visual sensitivity analysis (VSA), a branch of visual parameter space analysis, employs interactive visualizations to support exploratory understanding of model behavior [1]. Global sensitivity analysis (GSA), rooted in the design of experiments, relies on quantitative techniques to measure how variations in model inputs contribute to variations in model outputs [2], [3].

Despite their shared goal, VSA and GSA have largely evolved as non-intersecting fields. VSA typically does not leverage techniques of GSA to guide or structure its visual encodings, even though some approaches incorporate alternative forms of quantification, such as clustering [4]. Conversely, GSA rarely integrates visual analysis into its workflow: it provides numerical measures of effect strength but offers little support for understanding the shape, heterogeneity, or context of input–output relationships [5]. As a result, practitioners often learn which inputs matter, but not how they matter.

We argue that this missed opportunity for joint quantitative–visual reasoning is a key factor behind the limited adoption of both VSA [6], [7] and GSA [8], [9] in applied modeling practice. For GSA, also the the perceived complexity of its methods is identified as one of its adoption barriers [9]. In this work, we bridge the gap between GSA and VSA to provide both practitioners and researchers with an integrated, visual-quantitative approach for reasoning about complex simulation models.

On the one hand, the integration of VSA and GSA enables a qualitatively higher level of usefulness than either approach alone, supporting the democratization of advanced sensitivity analysis for practitioners. For VSA, numeric sensitivity information directs attention toward the most decision-relevant aspects of the model, filtering out inputs and effects of negligible importance and thereby clarifying visual representations in high-dimensional parameter spaces. In this way, GSA serves as a navigational aid in the iterative process of model exploration.

On the other hand, for GSA, visual representations of simulation ensembles allow it to go beyond scalar importance rankings towards revealing varying importance of inputs across the output values, different shapes of effects, and non-monotonicity that proves critical for decision support [10].

This study integrates GSA and VSA in a non-programmer-friendly dashboard designed to democratize advanced sensitivity analysis. The method behind the dashboard, simulation decomposition or SimDec, employs variance-based sensitivity analysis and visual partitioning of the output distribution by the regions of the most important inputs. This study extends the previous work on the dashboard [11] by the two-output SimDec plot, which displays not only key input-output relationships but also the connections between two chosen outputs, thereby enabling a more thorough examination of model behavior. We demonstrate the usefulness of this method via four examples: (i) an urban dynamics model with independent inputs, (ii) a heat exchange engineering system with dependent inputs, (iii) a multi-criteria decision-making model for optimizing corporate investment, and (iv) a breast cancer dataset, in which the causal mechanisms are unknown. Furthermore, following the footsteps of earlier visualization tool publications [12]–[14], we run a survey to evaluate dashboard effectiveness in democratizing advanced sensitivity analysis.

II. Critical literature review

We review the current literature from both research branches, VSA and GSA focusing on the features of particular interest for this paper: (i) the combination of quantification of sensitivity and visualization of effects; (ii) mapping the effects to the output variable values to provide an direct actionable insight (e.g. in order to get to certain output range, adjust A and B in a specific way); (iii) global multi-variable analysis able to capture interactions; and finally, (iv) ability to jointly analyze several outputs.

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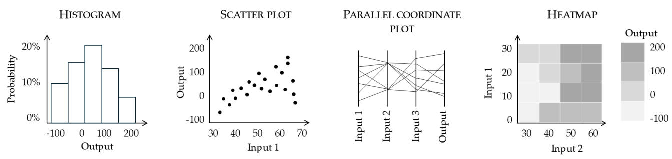


Fig. 1. Generic visualization types for sensitivity analysis.

A. Visual sensitivity analysis field

Visual parameter space analysis focuses on the analytical process of exploring and reasoning about simulation models, with specific visualization types serving as supporting tools rather than ends in themselves. Sedlmair et al. [1] established a unifying framework for parameter space analysis, which distinguishes different user navigation strategies and modeling tasks, including optimization, partitioning, fitting, outliers investigation, uncertainty, and sensitivity assessment. Visual sensitivity analysis, in particular, aims at answering the question of *'which changes of an output to expect with changes of inputs?'* and often used conjointly with other analysis tasks.

Visual sensitivity analysis scholars use a mix of generic (Fig. 1) and custom visualization types. Amirkhanov et al. [15] employ custom stability widgets, a color-coded parameter-space visualization with directional robustness indicators resembling heatmaps, and histograms. Histograms are a prevalent method for depicting Monte Carlo simulation results in many other studies, as well as software packages. Although helpful in illustrating the distribution of outputs, histograms alone do not establish any explicit connection to the input variables.

Berger et al. [16] opt for a combination of scatter plots and parallel coordinates plots. Scatterplots can depict individual input–output relations, but they possess a significant limitation: numerous plots are required to systematically explore all input variables, with no capacity to identify the higher-order interaction effects [17]. Several inputs can be included in a single scatterplot by utilizing different shapes and colors [18]. However, the unequal treatment of different inputs (X-axis versus shape) may bias the perception, and the graph quickly becomes messy. Parallel coordinates plots are proposed in some sensitivity packages [19]. However, the number of axes for multi-variable problems, if unfiltered, can confuse the overall perception, and the order in which axes are arranged can distort the perception of dependencies. For instance, as demonstrated in Fig. 1, the perceived influence of $Input2$ on the $Output$ may be impeded if $Input3$ intervenes spatially between them. This ordering bias raises concerns about the interpretative validity of parallel coordinate plots.

Matkovic et al. [20] employs a series of heatmaps, calling it families of data surfaces. Indeed, heatmaps are limited to two input variables at a time and therefore multidimensional problems require many of them. Even then, heatmaps represent only slices of hypercube, leaving large chunks of input-output space untouched. Thus, heatmaps can only assist in local sensitivity analysis. Evers et al. [21] use a combination of parallel coordinate plots and heatmaps for spatial sensitivity analysis. This work stands out by employing GSA to inform its visualization. Various other visual aids have been used in VSA, including response surfaces [22], marginal and conditional effect plots such as partial dependence and individual conditional expectation plots [23], [24].

Some recent VSA studies develop unique visualization approaches. Piccolotto et al. [25] introduce hierarchical clustering approach that portrays model's sensitivity structure as a dendrogram. Dendrograms allow analysts identify locally stable and sensitive parameter regions without requiring scalar features or explicit model structure. This makes the method broadly applicable to complex data types, including spatial and structured outputs. However, the results depend on clustering and normalization choices, and sensitivity is conveyed indirectly through cluster behavior rather than explicit input–output relationships. The approach is group-centric, which limits reasoning about dominance, trade-offs, and marginal effects, and it does not naturally support joint analysis of multiple continuous outputs.

Bauer et al. [4] adopt Voronoi cells to characterize sensitivity in terms of transitions between categorical (labeled) output states. This allows analysts to identify decision boundaries and regions where small input changes lead to switches between outcome classes, providing an intuitive view of local sensitivity without computing explicit sensitivity indices. However, the approach relies on discrete or discretized outputs and is therefore not applicable to continuous response variables. Sensitivity is expressed only as category switching, which limits analysis of effect magnitude, joint influences of multiple inputs, and trade-offs across several outputs. As a result, the method is best suited for classification-style models rather than general simulation-based sensitivity analysis. Interestingly, the authors considered incorporating global sensitivity analysis (GSA) but ultimately ruled it out, citing its inapplicability to categorical outputs. While it is true that some classical GSA methods assume continuous and differentiable model responses, this limitation is not universal. In particular, the efficient binning-based estimator for Sobol' indices employed in this paper [5] can accommodate such cases, and regional sensitivity analysis [26] is explicitly designed to handle categorical or threshold-based outputs.

B. Global sensitivity analysis field

Global sensitivity analysis (GSA) is a fundamental technique for enhancing the transparency, defendability, and trustworthiness of computational models. It enables analysts to identify and understand how uncertainty in input parameters affects output uncertainty, highlighting the most influential assumptions that require closer review. However, despite its conceptual significance and increasing methodological sophistication, the relative adoption rate of GSA across scientific and practical fields remains sporadic. Many studies indicate that numerous disciplines either ignore sensitivity analysis altogether or depend solely upon basic methods, like one-at-a-time (OAT) techniques, which do not explore the space mapped by the input-output relationship in full [9], [26]–[30].

There currently exists no universally accepted standard for visualizing results in GSA [5]. Even generic visualizations portrayed in Figure 1 with their limitations described above are only sporadically used in GSA studies. In practice, GSA outputs are frequently presented numerically, or through bar charts or box plots that display only the amplitude of sensitivity indices [31], [32]. Such an approach fails to reveal the functional form of input–output relationships. Reviews of GSA methods stress the need for enhancing interpretability and exposing complex interaction patterns [27], [33].

Rare attempts of VSA to enter the world of GSA remain within the realm of visualizing the strength of the relationships but not their shape. Fruth et al. [34] introduce FANOVA graphs for Sobol' indices, constructs with nodes representing input variables, edges representing pairwise total interaction indices, which aggregate higher-order Sobol' terms, and the thickness indicating the strength of the effects. Yang et al. [35] increase the level of visual sophistication for Sobol' indices proposing SenVis, an interactive visual analysis that uses tensor decompositions to enable scalable exploration of variable importance. While these frameworks facilitate the perception and exploration of sensitivity indices, they do not convey information about the shape or heterogeneity of input–output effects, their mapping to output values, or relationships across multiple outputs.

Another visualization approach, inspired by explainable artificial intelligence and Shapley values, computes sensitivity indices not only at the dataset level but also locally for individual data points, enabling visualization of how effect strength varies across the input or output space [36]. While this approach captures regional and heterogeneous variations in importance, it does not reveal the shape of the underlying input–output relationships or provide an explicit mapping from inputs to output values, nor does it support joint exploration of multiple outputs.

The Simulation Decomposition (SimDec) method, extended in this paper, maps input effects directly onto the distribution of output values, providing intuitive and actionable insights for modelers and decision-makers [10]. Leveraging sensitivity indices, SimDec partitions simulation outputs into subgroups defined by combinations of the most influential input variables. These subgroups are encoded as color-coded segments in stacked histograms, enabling simultaneous inspection of individual and joint input effects. While stacked histograms are a widely used visualization technique, they are typically employed to map a single categorical variable to an output distribution. In contrast, SimDec supports both categorical and continuous inputs and enables the concurrent mapping of multiple variables by constructing multi-variable scenarios—an extension that is non-trivial and becomes feasible only through an explicit algorithmic formulation.

C. Evolution of open-source and commercial software for global sensitivity analysis

The landscape of open-source and commercial software for GSA has grown considerably. This growth now provides researchers with diverse solutions across various programming paradigms to tackle computational and interpretability challenges in scientific and engineering applications. Python has become the dominant language, and SALib has established itself as the standard in this field. Its user-friendly implementations of Sobol' indices and Morris screening, along with comprehensive documentation and an active community, make it especially useful for educational and research purposes. With over 2 million downloads daily, integrating Sobol' index computation into SciPy has greatly increased GSA's accessibility within the wider scientific Python community.

Despite Python's dominance, R also maintains a strong presence amongst statistics-focused practitioners. The corresponding sensitivity package provides comprehensive coverage of GSA techniques, with particular strengths in handling dependent input variables [37].

Julia implementations such as GlobalSensitivity.jl leverage the language's just-in-time compilation for computational efficiency [38]. However, for most GSA applications, the computational advantages of compiled languages often prove secondary to workflow integration considerations, even for those cases involving millions of samples and multi-dimensional outputs. Consequently, interpreted languages typically suffice for analysis demands.

Specialized uncertainty quantification frameworks address methodological gaps left by general-purpose tools. UQpy offers a modular architecture enabling customized workflows through combinations of sampling strategies and sensitivity analysis methods. For high-dimensional problems, OpenTURNS provides sophisticated sampling and sensitivity analysis capabilities, though its implementation complexity can present barriers for non-specialist users. Institutional solutions further complement this ecosystem: Dakota (Sandia National Laboratories) delivers enterprise-grade capabilities in optimization-based sensitivity analysis and surrogate modeling [39], while PSUADE (Lawrence Livermore National Laboratory) provides rigorously validated implementations across nuclear engineering and climate modeling applications [40].

Newer, emerging paradigms have focused on democratizing access to GSA methodologies. Web-based platforms like SimDec have replaced programming requirements with intuitive visual interfaces, making sensitivity analysis more broadly accessible to policy analysts and domain experts without sophisticated computational backgrounds [41]. Cloud-based solutions such as UQLab Cloud and Dakota Cloud extend advanced analysis capabilities to researchers in resource-constrained environments, including institutions in developing countries.

Integration capabilities have improved substantially to reduce historical workflow barriers. Tools like pyDOE facilitate simulation software integration. Concurrent methodological innovations include sensobol's implementation of Jansen estimators with enhanced visualization, Morris' efficient screening for high-dimensional problems, and various emerging packages addressing dependent input challenges.

Critical limitations persist despite these advances. High-dimensional problems continue to require substantial computational resources that serve to exclude researchers without access to high-performance computing infrastructure. Steep learning curves for advanced methods necessitate extensive training, and fragmented toolchains force researchers to integrate disparate packages manually. Furthermore, comprehensive solutions that combine multiple techniques with robust visualization capabilities remain scarce. These challenges underscore the need for the continued development of accessible, integrated GSA frameworks that can balance computational efficiency with methodological rigor across diverse research communities.

III. Proposed approach: web dashboard for simulation decomposition

The open-source web dashboard for sensitivity analysis is built on Python [42] and is available at simdec.io. It operates based on variance-based sensitivity analysis with a simple binning method and a follow-up visual decomposition approach. Both are outlined in this section together with a description of the method's user interface.

A. Simple binning approach for Sobol' indices estimation

Variance-based GSA, primarily through Sobol' indices, is commonly used to measure how much each input factor affects the variance of a model's output [2]. The popular estimator [43], [44] allows for the calculation of these indices but has notable limitations: it requires a specifically structured sampling, demands direct access to the model for repeated simulations, and necessitates numerous runs to produce stable estimates. Additionally, it assumes input independence, which makes it unsuitable for applications with dependent or constrained input variables.

To address these challenges, several researchers have explored streamlined alternatives that rely on the post hoc analysis of existing simulation data. A key idea is to discretize or bin the input space, and compute the variance of the average model output within each bin [10], [45], [46]. This algorithm, implemented in the SimDec dashboard, has been shown to perform well even on highly non-rectangular or constrained input spaces [41], achieving good convergence even from simple random sampling and modest dataset sizes [5]. This makes it well-suited for analyzing both input-output data and non-simulated data.

By default, the dashboard displays the combined (fair Shapley allocation rule) indices, $S^{c}_{X_i}$, which is a sum of a variable's individual contribution (first-order effect), S_{X_i}, and half of its interaction effects (second-order effects) with all other inputs, $S_{X_iX_j}$.

$$S_{X_i} = \frac{\mathrm{Var}\left(\mathbb{E}(Y \mid X_i)\right)}{\mathrm{Var}(Y)} \tag{1}$$

$$S_{X_iX_j} = \frac{\mathrm{Var}\left(\mathbb{E}(Y \mid X_i, X_j)\right)}{\mathrm{Var}(Y)} - S_{X_i} - S_{X_j} \tag{2}$$

$$S^{c}_{X_i} = S_{X_i} + \sum_{\substack{j=1 \\ i\neq j}}^{K_{\text{inputs}}} \frac{S_{X_iX_j}}{2} \tag{3}$$

Barring sampling noise, the effects due to skewed factors distribution, and the presence of interaction terms higher than second order, the sum of the combined indices of all variables is equal to, or close to, one. In our experience, negative second-order indices resulting from dependent inputs compensate for their individual effects, and the sum of the combined indices remains close to one even in simple models with dependent inputs. However, for empirical datasets or models with complex input dependencies in which many of the second-order effects are negative, the dashboard described in the next section will display the first-order effects instead. These are used to navigate the follow-up decomposition. Numeric details for all indices, including first- and second-order indices, as well as combined indices, are available in the supplementary file that can be downloaded from the dashboard.

B. Simulation decomposition for visualization of input-output relationships

SimDec regroups the data into discrete input scenarios and visualizes the resulting output sub-distributions side-by-side [10]. This allows users to observe how different regions of the input space contribute to the output behavior, which reveals interactions, thresholds, and heterogeneous effects. The steps of the SimDec algorithm are straightforward, the elaboration for choices at each stage is discussed below:

1) Selection of input variables for decomposition (most important by default or user-specified)
2) Creation of 2–3 states for selected inputs by dividing their numeric ranges into equal parts
3) Formation of scenarios by combining all states of different inputs
4) Mapping the scenarios to the simulation runs or data points
5) Visualization of the decomposition using either a stacked histogram or a boxplot following a specific coloring logic [47]

Employing global sensitivity analysis at the first step allows mitigating the scalability limitation of the approach [48]. With the increasing number of variables chosen for decomposition, the number of scenarios increases, making the visualization less and less readable. However, there is a known Pareto principle in system modeling regardless of the application field, stating that the majority of system outputs are driven by the minority of inputs. Thus, identifying a few most important inputs eliminates the very need to display the effects of multiple input variables. In fact, in our series of experiments with several modelling teams from different academic fields collected in a book [41], in 85% of cases only three or less variables contributed to the majority of output variance, while their models often exceeded 10 input variables in total.

The variance-based approach to input importance quantification is chosen because it directly aligns with the visual effects of the decomposition: the higher the variance-based index, the further apart are the output sub-distributions resulting from different states of that input. And as described in the previous section, the classic variance-based indices estimator [43], [44] was modified to be able to work with a given data without the access to original model [5].

The second step further tackles the scalability issue by limiting the amount of states for each variable. Numerous experiments showed that when only two variables are chosen for decomposition, three states for each variable are often optimal, resulting in 9 scenarios. Whereas if three or more are chosen, the number of states is kept at 2 to decrease the number of scenarios. That, of course, only concerns continuous variables. If a categorical variable with 5 or less categories is chosen for decomposition, then each category becomes a state.

Third and fourth steps are the core of the decomposition algorithm and do not have any degrees of freedom.

The last fifth step involves visualization and has design choices in the form of the visualization and the coloring logic. The coloring logic was settled after numerous experiments: the sub-distributions of the first in decomposition variable are assigned main distinct colors and all other subdivisions get the shades of the main colors produced by changing the brightness of main colors at consistent intervals defined based on the number of scenarios [47]. Such approach enables easier perception of input interactions, where the effects of one input can be contrasted in different states of another input through different behavior of shades of the main colors.

As for the visualization form, the histogram was initially chosen, as it is a classis approach to visualize results of a Monte Carlo simulation used by scholars across the fields. The choice between overlay and stacked histograms was studied separately and favored stacked ones for several reasons, including preserving the shape of the original full distribution and no data loss due to overlaying, even though at a cost of the shape perception of sub-distributions [49]. Nevertheless, it was later shown that it is not the shape of the sub-distributions but their interposition is what matters for reading heterogeneous effects in complex models [10].

Stacked histogram, however, becomes poorly readable for cases with highly skewed distributions or little data in some scenarios due to dependencies. Thus, a boxplot visualization was introduced to tackle this problem [11]. The same decomposition and coloring algorithms are applied to box plots, only instead of series of a stacked histogram, every scenario is visualized as a box plot.

Finally, models with several outputs raised another visualization challenge. Even though each output can be represented with its own decomposition, such approach does not allow tracing simulation results across the outputs, unless outputs are highly correlated and produce exact same decomposition. Meanwhile, investigating cross-output effects is important for decision support in a multiple targets problem. Therefore, in this paper, we introduce a two-output plot, which is a scatter plot with histograms with applied to it decomposition algorithm. And while a scatter plot with histograms has been widely used [50], when it is decomposed into series, they are predominantly formed by categories of a single variable. Whereas combining it with SimDec algorithm allows mapping several continuous variables onto the output distribution.

C. User interface

The dashboard implements both algorithms, conjointly – the computation of Sobol' indices via the simple binning approach and the consecutive visual decomposition guided by

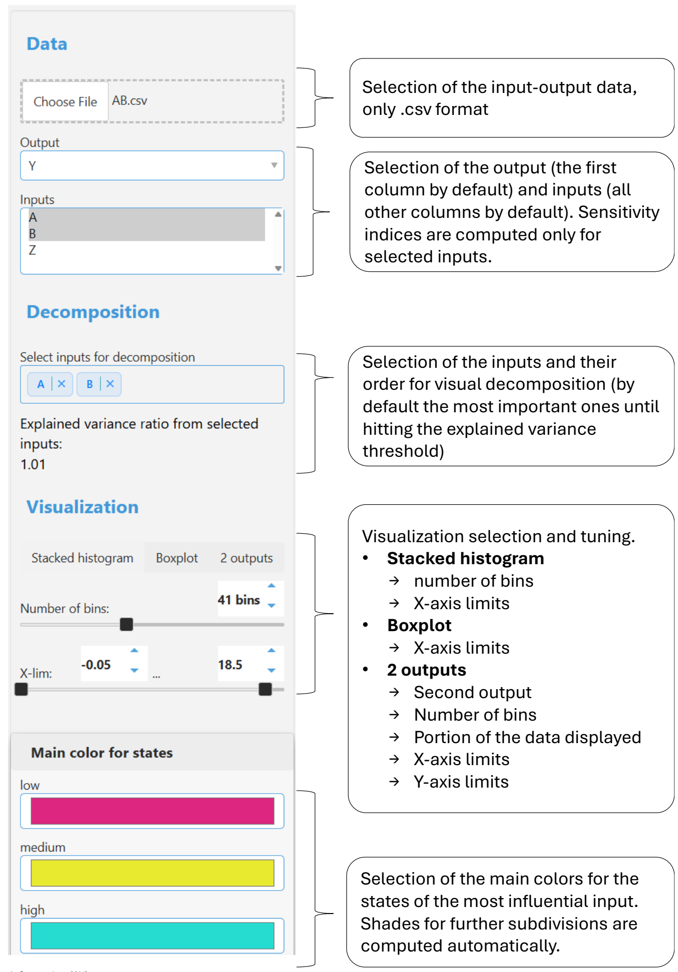


Fig. 2. The control panel for the SimDec dashboard.

the computed indices. The entire process occurs automatically upon the uploading of a new dataset and can be further customized by the user, see Fig. 2.

The left panel allows the user to make the following choices:

- Set of inputs to be selected, important when the model contains multiple non-influential inputs that can be omitted from the analysis
- Type of visualization, as described subsequently

To illustrate the dashboard functioning, we create a simple model with two inputs and two outputs:

$$Y = A + B \tag{4}$$

$$Z = AB \tag{5}$$

where A and B are uniformly distributed between (0, 10) and (0, 8), respectively. The model is run ten thousand times, and the resulting input-output dataset is loaded onto the dashboard. After removing output Z from the list of inputs, the dashboard automatically performs decomposition and displays the results in the default stacked histogram visualization, Fig. 3.

In Fig. 3, the dashboard presents sensitivity indices (bottom left), details on how the selected inputs were broken down

Fig. 3. Default dashboard results for the simple model.

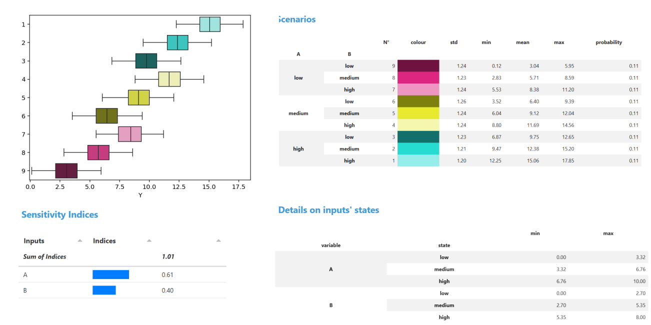


Fig. 4. Dashboard results for the simple model visualized with boxplots.

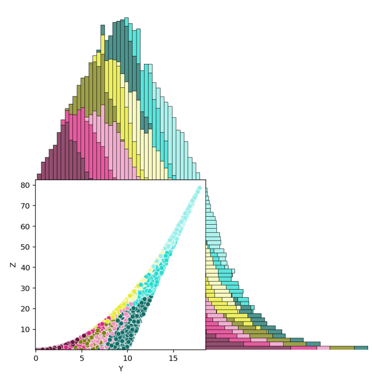


Fig. 5. Plots for Y and Z relative to the same decomposition shown in Fig. 3.

into states (bottom right), the formation of scenarios from all combinations of states with their corresponding statistics (top right), and the resulting visualization (top left).

The same decomposition can also be presented using boxplots, while everything else remains unchanged, Fig. 4.

If, as in the present example, there is a second output to consider, the relationship between the two outputs is generated, see Fig. 5.

The dots in the scatter plot and their colors correspond to the distributions shown at the top for the first output and on the right for the second. This enables a detailed, visual cross-comparison of the same decomposition from three very different perspectives.

D. User survey

The dashboard survey was conducted to explore how convenient the dashboard is for practitioners, whether it provides an additional insight into model behavior, and, more broadly, to collect feedback, identify major issues, and highlight areas for improvement. The survey consisted of four blocks: user experience, model complexity, dashboard handling, and dashboard value. It included Likert-scale evaluations (1–5), multiple-choice, and open-ended questions. The results are summarized in the Supplementary material.

The sample of 21 respondents reported below-average programming experience and limited familiarity with global sensitivity analysis. The practitioners' models varied in both complexity and domain; on average, they had six uncertain inputs and, in more than half of the cases, a single output.

The overall experience with dashboard handling was rated 4 out of 5. Most users watched the dashboard tutorial and encountered no technical issues.

The perceived value of the dashboard was also rated 4 out of 5. Respondents generally agreed with statements such as "The dashboard helped me understand how inputs affect outputs in my model," "This dashboard lowers the barrier to sensitivity analysis compared to script-based tools," and "I would like to use this dashboard for future analyses."

The reported technical issues and suggestions informed further improvements to the dashboard. One recurring issue was that the dashboard would load indefinitely, which was traced to incorrect input data formats. This was addressed by implementing error messages that guide users to specific problems in their datasets. Another concern related to slow loading times was linked to cloud deployment; this was resolved by moving the dashboard to run entirely on the client side.

Overall, the user survey results indicate that the dashboard makes comprehensive global and visual sensitivity analysis accessible to practitioners with limited programming experience and no prior background in sensitivity analysis.

IV. Cases

A. Model with independent inputs

This case is based on the URBAN1 model, a simplified system dynamics model adapted by [51] from Jay Forrester's original Urban Dynamics framework [52]. The model captures the endogenous interactions between three key urban subsystems: population, housing, and business structures. It simulates how feedback loops within and across these sectors drive urban growth, stagnation, or decay. Despite its simplicity, the model reproduces typical urban phenomena like employment imbalances, housing shortages, and infrastructure constraints. It serves as an effective tool for illustrating how well-intentioned policies can lead to counterintuitive or even adverse outcomes if systemic interactions are not fully understood [53].

The model is reproduced in a spreadsheet environment, and the four key inputs are varied, as shown in Table I.

Population is identified as the main output of the model, and its measured value is averaged over the last 4 years of the model's 100-year time horizon to diminish its cyclicality

TABLE I
INPUT VARIABILITY FOR THE URBAN1 MODEL

Input	Distribution
Immigration rate	$U(0.05, 0.20)$
Outmigration rate	$U(0.05, 0.20)$
Birth rate	$U(0.02, 0.05)$
Death rate	$U(0.01, 0.02)$

effect. The second output of interest is the Labor force to jobs ratio, which signals unemployment if it is above one and an insufficient labor force if it is below.

The model is simulated ten thousand times, and the input-output data is uploaded to the dashboard. After removing the Labor force to jobs ratio from the list of inputs, the dashboard displays the result shown in Fig. 6.

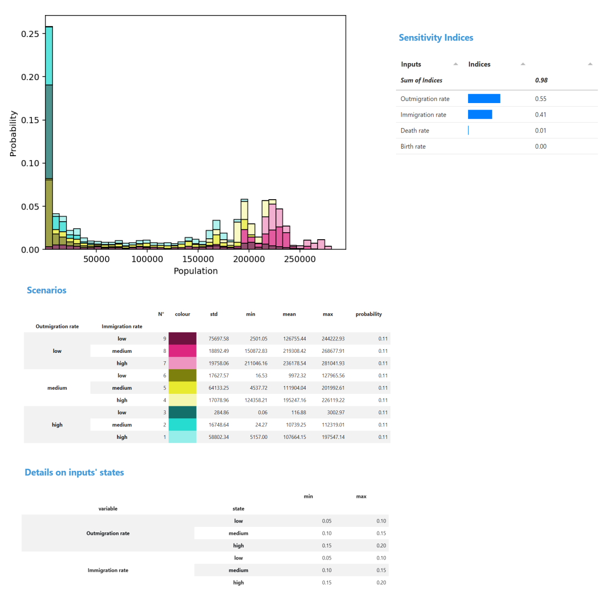


Fig. 6. Dashboard results for the URBAN1 model.

The model exhibits high sensitivity to migration flows, which dominate urban population dynamics in this structure. Differences in output distributions across scenarios reveal strong nonlinear effects, with some configurations locking the system into decay. The influence of this factor can be observed in the spike of outputs below the initial population level of 50,000 people. The scenarios have equal likelihood, which confirms the independence of the inputs. The most prosperous population (red scenarios) results from low outmigration and high immigration rates. In contrast, urban decay can occur in various scenarios involving high and medium outmigration, as well as medium- and low-immigration.

Bringing a second output into the picture - not as a separate analysis but as part of the same decomposition - can provide deeper insights into the system's behavior, as shown in Fig. 7.

The two-output visualization reveals a strong, nearly-linear, positive correlation: as the population increases, the labor-to-jobs ratio also rises. However, this suggests a growing mismatch between labor supply and job availability in larger

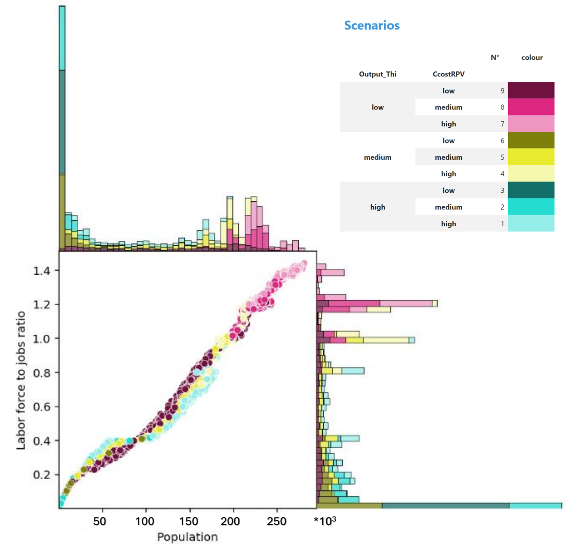


Fig. 7. Two-output plot for population and labor force to jobs ratio of the URBAN1 model.

populations, which would signify rising unemployment as the system struggles to adjust to higher immigration rates. The previously prosperous system indicates signs of distress, and a more balanced state becomes evident at population levels around 200,000.

The light blue scenario, which represents high outmigration and immigration, exhibits a distinct S-shaped pattern in the scatter plot, which differs considerably from the mostly linear trends observed in the other scenarios. This pattern suggests that the relationship between population size and the labor-force-to-jobs ratio in this scenario is nonlinear and regime-shifting. These complexities would not become readily apparent without the visual observations revealed in the two-output plot.

B. Model with dependent inputs

We use the optimization model of [54] to illustrate a case containing dependent input factors. The model involves the techno-economic optimization of a primary heat exchanger and pressure vessel in a small modular nuclear reactor (SMR) designed for district heating. The optimization is performed based on the Cuckoo Search metaheuristic algorithm [55]. The optimization aims to minimize the Levelized Cost Of Heat (LCOH) by adjusting the design parameters in response to cost uncertainty and variability in thermohydraulic and mechanical conditions, as shown in Table II.

The optimized design characteristics can be considered simultaneously as outputs from the optimization or as dependent inputs that shape the ultimate output – LCOH. Of the many possible angles examined by [54], we choose to demonstrate the decomposition of the optimized length of tubes, Fig. 8. The two most influential factors appear to be a mix of dependent (optimized primary water inlet temperature) and independent (RPV cost coefficient) variables.

Fig. 8 displays a histogram with four peaks or regions around which the optimization tends to gravitate. The first two are predominantly produced from high- and medium-inlet temperatures and low- and medium-costs. The third peak is dominated by medium values of both variables, and the fourth one by low-inlet temperatures and high costs, excluding high-inlet temperatures entirely. Nevertheless, each peak can be produced from a variety of combinations of variable values. Similarly, numerous optimization results are found between peaks, resulting from different scenarios. This diverse spectrum of results can be attributed to the influence of other variables in the model and to the heuristic nature of the optimization engine. The graph illustrates that changing the input values can significantly impact the results of the optimization, leading, in particular, to structurally distinct designs.

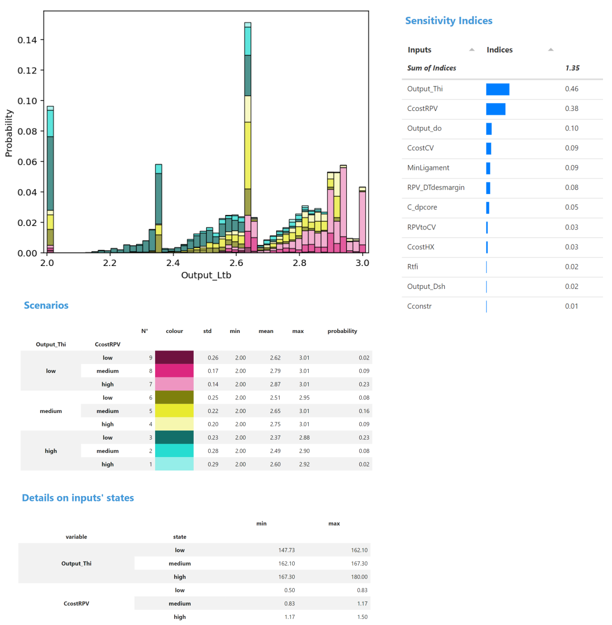


Fig. 8. SimDec dashboard results for the optimized tube length.

The SimDec visualizations further reveal the inner workings of the optimization logic. Combinations of low-temperature and low-costs, as well as high-temperature and high-costs, are rare (see the lower probabilities of the corresponding first and last scenarios of the decomposition, Fig. 8). Different cost levels, treated by the model as external uncertainty, lead to different heat exchanger designs: high-costs result in an optimized design with thinner RPV walls that withstand lower-pressure requiring lower-inlet temperatures and longer tubes (lighter shades in Figure X or scenarios 7 and 4), while low costs lead to a high-pressure, compact RPV with higher inlet temperatures and shorter tubes (darker shades in Fig. 8 or scenarios 3 and 6).

Interestingly, the SimDec revelations were unexpected by the modelers [54]. The default 6-meter tube length can be cut without scrap into 2- or 3-meter tubes. Since the model is minimizing the overall costs, any other tube lengths should be suboptimal. Nevertheless, the optimization not only returns numerous values that are not close to 2 or 3 (over 85% of the data) but also produces mysterious peaks in the middle of the range. An explanation related to certain other mechanical design limitations was eventually identified.

After the original analysis [54], a two-output visualization can be used to further investigate the model behavior by exploring how the different designs translate into the ultimate LCOH output values and how the critical mechanical design factors affect LCOH. By assigning the second output to LCOH

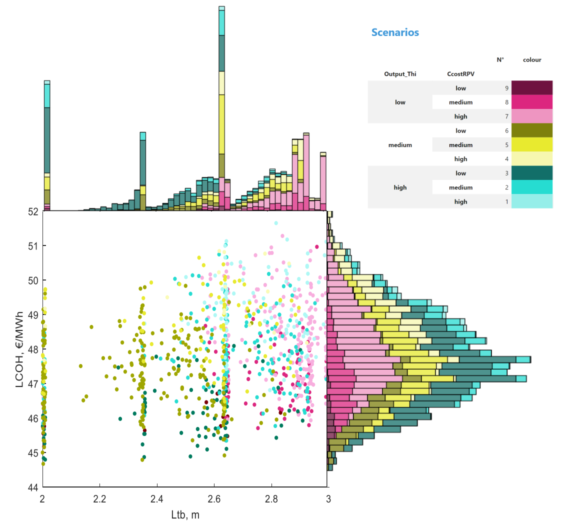


Fig. 9. Two-output plot for tube length and levelized cost of heat.

TABLE II
INPUT PARAMETERS FOR THE OPTIMIZATION MODEL

Variable	Explanation	Range
Uncertainty / variability: Thermo-hydraulic		
$R''_{tf,i}$	Thermal fouling resistance $(\mathrm{m^2K})/\mathrm{W}$	$[0, 8 \cdot 10^{-5}]$
$C_{\Delta p(\text{core})}$	Core pressure drop factor	$[0.5, 1.5]$
Uncertainty / variability: Mechanical		
Min. P-d_o	Ligament thickness (mm)	$[2.4, 4.0]$
$RPV\ \Delta T_{des}$	Thermal margin (°C)	$[20, 40]$
s_{RPV-CV}	RPV–CV spacing (m)	$[0.3, 1.0]$
Uncertainty / variability: Costs		
C_{constr}	Construction cost factor	$[1.5, 2.5]$
$C_{cost,HX}$	Heat exchanger cost factor	$[0.7, 1.3]$
$C_{cost,RPV}$	RPV cost factor	$[0.5, 1.5]$
$C_{cost,CV}$	CV cost factor	$[0.5, 1.5]$
Optimization results: Design parameters		
L_{tb}	Tube length (m)	$[2, 6]$
d_o	Tube diameter (mm)	$\{8, 10, 12, 14, 15\}$
D_{sh}	Shell diameter (mm)	$\{323.9, 355.6, 406.4, 457, 508\}$
T_{hi}	Inlet temperature (°C)	$[145, 180]$
Optimization results: Target		
$LCOH$	Levelized cost of heat (€/MWh)	–

in the dashboard, we can address these questions as shown in Fig. 9.

Although a considerable overlap can be observed, the scatterplot indicates that an increased tube length raises the LCOH. The overlap can only be partly explained by the graph, suggesting that other factors are at play for LCOH. We know from previous work [54] that 65% of LCOH variation is explained by CV cost not considered in this decomposition (as it does not affect the tube length). Focusing on the regions of LCOH, scenarios 6 and 9 (dark yellow and red) are the most attractive and keep the cost down compared to all other scenarios. These regions are formed by low RPV cost and below medium inlet temperatures. These observations correspond to the first three peaks of tube length at 2, 2.4, and 2.6.

For this dependent input case, the SimDec dashboard revealed multiple insights. It clearly demonstrates an absence of monotonicity in the optimized results (peaks of tube length and feasible solutions across the graph). The two-output visualization uncovers distinctively scenario-specific optima and exposes how RPV cost and inlet temperature jointly shape the preferred heat exchanger design. SimDec's unique visual combination of scatterplots between outputs, their histograms, and the input-scenario-based coloring allows an assessment of where the optimized results concentrate, which combinations yield cost-effective solutions, and how uncertainty and the design choices interact.

C. Multi-criteria decision-making model

A classic multi-criteria decision-making model, often taught in operations research, considers a firm that must decide on three variables x1,x2, and x3, which are the level of production of three goods, while pursuing goals related to profit, investment, and employment [56]. The firm seeks to achieve a profit exceeding 125 million dollars, maintain employment at 4 hundred employees, and limit investment to 55 million dollars. The relationships among the decision variables (the products) are specified as follows:

$$12x_1 + 9x_2 + 15x_3 \geq 125 \quad (6)$$

$$5x_1 + 3x_2 + 4x_3 = 40 \quad (7)$$

$$5x_1 + 7x_2 + 8x_3 \leq 55 \quad (8)$$

Constraints 1 and 3 are given in million dollars, and constraint 2 is given in hundred employees. Penalty weights are assigned to deviations from the stated goals: 5 for each unit below the profit target, 3 for each unit above the investment limit, 4 for each unit exceeding the employment cap, and 2 for each unit falling short of the employment requirement. The problem is then formulated as minimizing the overall penalty Z, under the assumption that penalties are additive.

From the perspective of sensitivity analysis, this problem can be approached in two distinct ways. On the one hand, the optimized solution itself can be subjected to sensitivity analysis (as in the previous case), whereby each simulation run yields a new optimal solution. On the other hand, one can examine the solution space of the problem and study how it responds to the introduction of uncertainties. Since the former approach has already been demonstrated, the focus here is on exploring the latter.

First, to explore the solution space of the deterministically formulated problem, the decision variables are assigned possible ranges from 0 to 10, and a Monte Carlo simulation is run to compute the overall penalty, denoted as $Z_{d}eter_{p}rof$. Second, uncertainty is introduced into the profit assumptions. Instead of the fixed coefficients of 12, 9, and 15, uncertain ranges are applied of [10, 12], [8, 9], and [5, 20] in the first equation governing profit. Ceteris paribus, the overall penalty $Z_{u}nc_{p}rof$ is then computed within the same simulation and contrasted with its deterministic counterpart. Fig. 10 presents the resulting two-output decomposition.

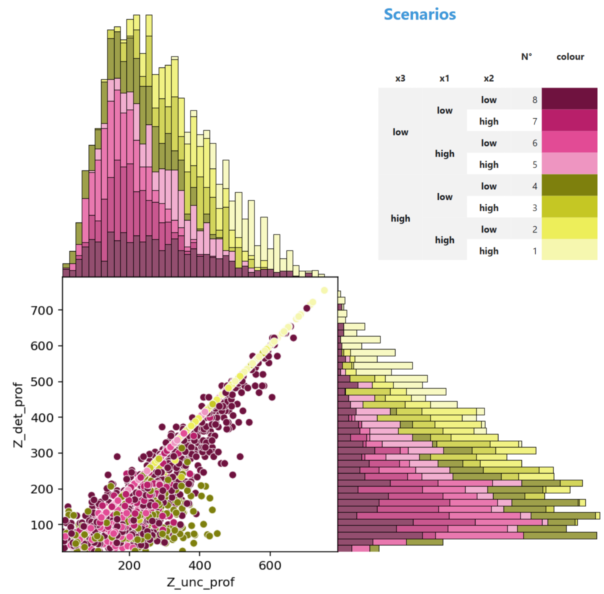


Fig. 10. Two-output plot for penalty with (right) and without (top) profit uncertainty.

The algorithm identified the following order of importance among the decision variables: $x3$, $x1$, $x2$. All three are used to decompose the distributions of penalties. More desirable solutions lie toward the left (and bottom) of the plots, where the penalty is minimal, corresponding to lower quantities of all products. What is particularly interesting, however, is the difference between the penalties with- and without-profit uncertainty. The scatterplot enables us to examine this difference in detail visually.

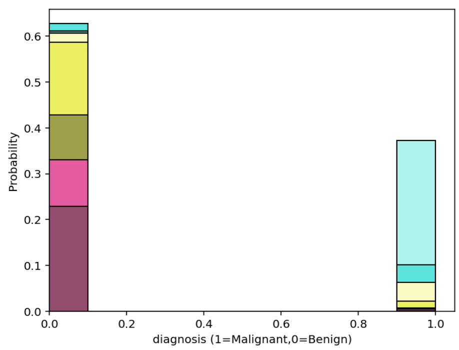
A perfect correlation (the diagonal line) emerges from the scatterplot cloud at the point where the achieved profit meets the goal (zero penalty). This line is formed by various combinations of products (represented by different colors), but it is dominated by solutions with high values of $x3$ and $x1$ (lighter shaded yellow dots). If profit uncertainty had no effect, all points would lie neatly along this diagonal.

However, the less organized cluster of dots near the optimal solution illustrates the distortions that profit uncertainty introduces into the solution space. As most points fall below the diagonal, this indicates that the penalties under uncertainty are higher than those in the deterministic case. Furthermore, the effect is clearly not uniform across portfolios, as the most significant deviations are observed for dark red solutions with all products low and for dark yellow solutions with high x3 and the other products low. A substantial share of the cloud consists of portfolios that appear favorable under the deterministic formulation with low penalties, yet lose credibility once uncertainty is introduced. The visual contrast shown in Fig. 10 highlights how uncertainty reshapes not only the magnitude of penalties but also the relative attractiveness of different product mixes.

D. Empirical breast cancer diagnostic data

An empirical breast cancer dataset is used in this section to illustrate a model-free use case for the dashboard [57], [58]. The data contains a binary output variable representing the diagnosis and thirty characteristics (radius, texture, perimeter, concavity, etc.) of the breast mass image computed for each cell nucleus.

Loading the dataset into the dashboard produces the following result, Fig. 11.



Scenarios

concave points_worst	perimeter_worst	N°	colour	std	min	mean	max	probability
low	low	9		0.09	0.00	0.01	1.00	0.23
	medium	8		0.18	0.00	0.03	1.00	0.11
	high	7		nan	nan	nan	nan	0.00
medium	low	6		0.13	0.00	0.02	1.00	0.10
	medium	5		0.29	0.00	0.09	1.00	0.17
	high	4		0.47	0.00	0.68	1.00	0.06
high	low	3		0.00	0.00	0.00	0.00	0.01
	medium	2		0.46	0.00	0.71	1.00	0.05
	high	1		0.00	1.00	1.00	1.00	0.27

Fig. 11. SimDec dashboard results for the breast cancer dataset.

The binary diagnosis variable is represented naturally by only two bars and decomposed by the two most influential explanatory variables – concave points and perimeter. While a stacked histogram with only two bars might not provide the aesthetically best presentation, we can already detect a good visual separation between input values in the output distribution. Most cases with low and medium concave points (red and yellow shades) fall into the benign diagnosis (output = 0), while cases with high concave points (blue) form the malignant diagnosis. There are a few borderline cases that appear in both diagnoses. These are formed by a combination of either medium concave points and a high perimeter (light yellow) or high concave points and a medium perimeter.

For such data with binary or categorical outputs, the two-output plot of the dashboard can be used to visualize the relationship in an alternative, novel way. Namely, the two most influential inputs can be displayed on axes instead of outputs, while the binary output categories would serve as the decomposition variable. Such a graphical decomposition projection of the output variable onto the influential inputs is illustrated in Fig. 12.

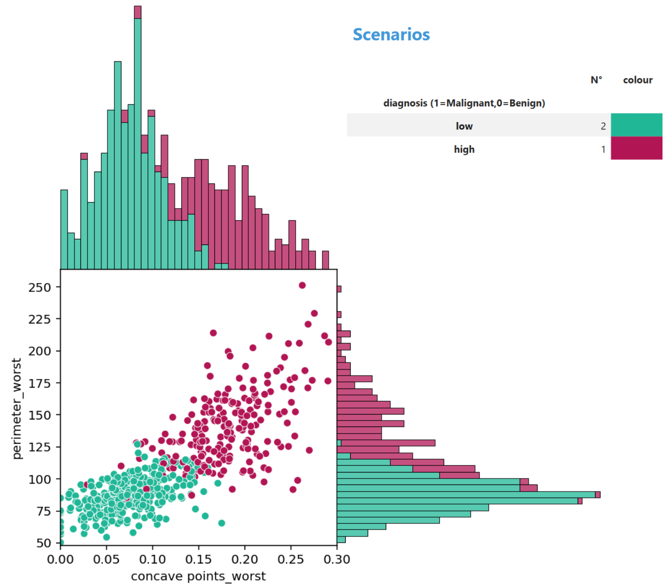


Fig. 12. An alternative way to use a two-output plot: the two most influential inputs are displayed on axes, while the output's categories are used to decompose the data.

While this alternate graph provides a more readable and intuitive picture of the empirical data, identical diagnostic conclusions to those from Figure 11 can also be derived from this projection. The data possesses several other explanatory variables with high sensitivity indices, and the dashboard can be readily employed to provide a visual exploratory analysis of their relative impacts. In this sense, the dashboard can also function as a convenient, quantitatively-guided tool for performing interactive, visual explorations of data. However, it is not intended as a substitute for predictive analytics.

V. Discussion and Conclusions

Frequently, the choice of optimization versus exploration is driven primarily by culture - both disciplinary and political - as the techniques for quantification are generally not neutral [59]. For example, in impact assessment studies, an efficiency-driven logic would point to the use of optimization, where a crisp number is preferred as the anchoring device for policy-relevant studies [59]. Crisp, precise numbers create an impression of prediction and control [60]. A precautionary mindset may lend itself to a thorough exploration of the uncertainty space, including those driven by the choice of the options [61], [62]. Some researchers in the operations research domain have embraced sensitivity analysis by reconnecting to the concept of tolerance [63].

There is a growing call in the literature to move beyond narrowly-framed, technically-sophisticated yet conceptually-limited models toward more explorative, reflexive approaches that can better support decision-making under uncertainty. Scoones [64] argues for embracing plural, adaptive, and context-sensitive models to navigate uncertainty in complex socio-ecological systems, while Saltelli et al. [65] emphasize the need for sensitivity analysis to foster humility, transparency, and meaningful engagement with uncertainty, rather than reinforce overconfident quantification.

Though much has improved in the use of sensitivity analysis [3] — understood here as the practice of Monte Carlo-based global sensitivity analysis combined with uncertainty analysis (or uncertainty quantification) — many studies are still performed by altering only one factor at a time [8]. Practitioners of global sensitivity analysis consider this approach "perfunctory" [28], as it explores only a tiny fraction of the input factors' hyperspace.

It is perhaps a consequence of the highly specialized organization of science into separate disciplines, journals, and practices that the adoption of global sensitivity analysis remains fragmented across the various disciplines [9]. The entry barriers to the implementation of global sensitivity analysis remain high due to the complexity of its methods and the veritable paucity of its systematic teaching.

The web dashboard for global sensitivity analysis and interactive model exploration presented in this article provides a concrete solution: it empowers non-expert users to explore model behavior, test assumptions, and visualize uncertainty without the need for technical coding skills. This democratizes critical model analysis tools and supports broader, more inclusive, and reflective engagement with decision-relevant modeling aspects, which aligns with the vision proposed in the literature.

The dashboard generates new types of insights beyond accessibility by making the inherent sensitivity structure of models and data directly observable. Users can not only identify which specific variables drive variation in outcomes, but also visually examine the shapes of the input-output and output-output relationships, which often proves crucial for the decision [10]. In this way, the tool fosters an explorative mindset in modelling, and a more meaningful engagement with uncertainty — both of which are essential for modern decision support.

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Supplementary Material for Global Sensitivity Analysis Made Easy Through Interactive Visualization

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TABLE I
SUMMARY OF DASHBOARD SURVEY RESPONSES

Question	Summary of response
User experience	
How would you rate your programming experience?	2.86
How familiar are you with global sensitivity analysis (GSA)?	2.24
Model complexity	
Which best describes your model behavior?	
– mostly linear	38%
– nonlinear	19%
– contains thresholds/scenarios	19%
– I am not sure	24%
How many input variables are randomized/varied in your model?	$\mu = 6$
How many output variables are you tracing?	median 1
What's the model domain?	
– Economics/Finance	52%
– Engineering	19%
– Environment	14%
Dashboard handling	
Did you watch the video tutorial?	
– Yes—I needed it to use the dashboard	43%
– Yes—but I could have used the dashboard without it	29%
– No—I did not need the tutorial	29%
I found the dashboard easy to use	4.1
I would need technical assistance to use this dashboard	2.1
The dashboard was unnecessarily complex	1.95
Calculated experience grade	**4.02**
Did you encounter any technical issues?	
– No	67%
– Yes, please specify	33%
Which graphs did you use to explore your model?	
– Histogram	91%
– Boxplot	48%
– Two-output plot	43%
Perceived value of the dashboard	
The dashboard helped me understand how inputs affect outputs in my model	3.81
This dashboard lowers the barrier to sensitivity analysis compared to script-based tools	4.00
I would like to use this dashboard for future analyses	4.19
Calculated value grade	**4.00**